

15.2 Estimable contrasts

Most questions about the treatments of a one-way experiment are contrasts. Does fertilizer help at all? Is the highest rate worth its cost? Which pairs of treatments differ? Earlier chapters supplied the theory: estimability (Chapter 8), the t test and interval (Chapter 11, Chapter 12) and simultaneous methods (Chapter 13). This section collects them for the one-way analysis, adds a geometric fact about power, and applies them to the nitrogen trial.

15.2.1 Contrasts are the treatment space

Recall that a contrast is a linear function $\psi = \sum_k c_k \mu_k$ with $\sum_k c_k = 0$ and $\mathbf{c} \neq \mathbf{0}$ (Definition 8.7.1). In the effects form it equals $\sum_k c_k \alpha_k$, and the contrasts are exactly the estimable functions of the effects alone (Theorem 8.7.1). Their estimates are $\hat{\psi} = \sum_k c_k \bar{y}_k$, whatever side condition or coding was used to fit the model.

Behind this lies a simple geometry. Let

$$\mathcal{T} = \mathcal{C}(\mathbf{Z}) \cap \mathbf{1}^\perp$$

be the **treatment space**: the vectors that are constant within groups and sum to zero. Its dimension is $g - 1$ (Theorem 6.5.1). A vector $\mathbf{u} \in \mathcal{T}$ takes some value v_k on every observation of group k , with $\sum_k n_k v_k = 0$. Put $c_k = n_k v_k$. Then $\mathbf{c}$ is a contrast and $\mathbf{u}$ is the vector $\mathbf{u}_\mathbf{c} = \mathbf{Z}\mathbf{D}^{-1}\mathbf{c}$ of Eq. 8.7.2, with $\mathbf{D} = \text{diag}(n_1, \dots, n_g)$. Conversely every $\mathbf{u}_\mathbf{c}$ lies in $\mathcal{T}$ (Proposition 8.7.2). So contrasts and nonzero vectors of the treatment space correspond one to one, and $\hat{\psi} = \mathbf{u}_\mathbf{c}^\top \mathbf{y}$. *A contrast is a direction in the treatment space, and its estimate is the component of the data along that direction.*

Equal treatment means say that the mean vector $\boldsymbol{\theta} = \mathbf{E}(\mathbf{Y})$ has no component in $\mathcal{T}$, that is, that every contrast vanishes (Exercise (A2)). The overall test of Section 15.3 looks at all of $\mathcal{T}$; a single contrast looks along one line of it.

15.2.2 The test and interval for one contrast

For a contrast $\mathbf{c}$ write

$$v_\mathbf{c} = \sum_{k=1}^g \frac{c_k^2}{n_k},$$

so that $\text{Var}(\hat{\psi}) = \sigma^2 v_\mathbf{c}$ (Eq. 8.7.1).

Proposition 15.2.1. Sum of squares and test for a contrast

In the normal one-way model, let $\psi = \sum_k c_k \mu_k$ be a contrast, $\hat{\psi} = \sum_k c_k \bar{Y}_k$, and $\psi_0 \in \mathbb{R}$. Then:

- (a) $\hat{\psi} \sim \mathbf{N}(\psi, \sigma^2 v_\mathbf{c})$, and $\hat{\psi}$ is independent of s^2 ;
- (b) the statistic

$$T = \frac{\hat{\psi} - \psi_0}{s\sqrt{v_\mathbf{c}}} \sim t(n - g, \delta), \quad \delta = \frac{\psi - \psi_0}{\sigma\sqrt{v_\mathbf{c}}},$$

so the test of $\psi = \psi_0$ rejects when $|T| > t_{n-g, \alpha/2}$, and $\hat{\psi} \pm t_{n-g, \alpha/2} s\sqrt{v_\mathbf{c}}$ is an exact $1 - \alpha$ confidence interval for ψ ;

(c) the **sum of squares of the contrast**

$$SS(\mathbf{c}) = \frac{\hat{\psi}^2}{v_{\mathbf{c}}} = \|\mathbf{P}_{\mathbf{c}}\mathbf{Y}\|^2,$$

where $\mathbf{P}_{\mathbf{c}}$ projects onto the line spanned by $\mathbf{u}_{\mathbf{c}}$, has one degree of freedom, and $SS(\mathbf{c})/s^2 = T^2$ for $\psi_0 = 0$ has the $F(1, n - g, \delta^2)$ distribution. Replacing $\mathbf{c}$ by $a\mathbf{c}$ with $a \neq 0$ changes neither $SS(\mathbf{c})$ nor the test;

(d) $SS(\mathbf{c}) \leq \sum_k n_k (\bar{Y}_k - \bar{Y})^2$, with equality when $c_k = n_k(\bar{Y}_k - \bar{Y})$ for every k .

Proof. Use the cell-means form, with model matrix $\mathbf{Z}$ of full rank g , so that $(\mathbf{Z}^\top \mathbf{Z})^{-1} = \mathbf{D}^{-1}$ and $\mathbf{c}^\top \mathbf{D}^{-1} \mathbf{c} = v_{\mathbf{c}}$. The function $\mathbf{c}^\top \boldsymbol{\mu}$ is estimable with estimate $\mathbf{c}^\top \hat{\boldsymbol{\mu}} = \hat{\psi}$. (a) and (b) are Theorem 11.6.1(a) and Corollary 7.5.2 with $\boldsymbol{\lambda} = \mathbf{c}$, $r = g$, and the interval is Theorem 12.1.1. The independence of $\hat{\psi}$ and s^2 is part of Theorem 7.5.1. (c) The identity $\hat{\psi}^2/v_{\mathbf{c}} = \|\mathbf{P}_{\mathbf{c}}\mathbf{Y}\|^2$ is Proposition 8.7.2(b), and T^2 is the F statistic of the hypothesis $\psi = 0$ by Theorem 11.6.1(b), with noncentrality δ^2 . Scaling $\mathbf{c}$ by a multiplies $\hat{\psi}^2$ and $v_{\mathbf{c}}$ by a^2 , and leaves the line spanned by $\mathbf{u}_{\mathbf{c}}$ unchanged. (d) is Exercise (B1). ■

By part (c) a contrast's sum of squares is a projection onto a line of $\mathcal{T}$, and by part (d) no line captures more than the whole space. The line that captures all of it is chosen by the data, which is why its t statistic needs Scheffé's multiplier (Theorem 13.2.2).

15.2.3 Planned contrasts in the nitrogen trial

Before the trial the agronomist wrote down two questions. First, does nitrogen raise yield at all? The natural contrast compares the average of the four fertilized rates with the control, $\psi_1 = \frac{1}{4}(\mu_2 + \mu_3 + \mu_4 + \mu_5) - \mu_1$. Second, is the highest rate better than the moderate one? That is $\psi_2 = \mu_5 - \mu_3$, the difference between 160 and 80 kg N/ha.

Example. Two planned contrasts

The pooled standard deviation is $s = 0.5211$ on 25 degrees of freedom. For ψ_1 , $v_{\mathbf{c}} = \frac{1}{6}(1 + 4 \cdot \frac{1}{16})$, the estimate is 2.110 t/ha with standard error 0.238, so $t = 8.87$ and the 95% interval is [1.620, 2.600]. Fertilizer raises yield by roughly two tonnes per hectare. Its sum of squares is 21.370.

For ψ_2 the estimate is 0.595 t/ha with standard error 0.301, $t = 1.98$, two-sided $p = 0.059$, and the interval [−0.025, 1.215]. The data are consistent with anything from no gain to a gain of more than a tonne: with six plots per rate, half a tonne is only about 1.7 standard errors of a pairwise difference.

```
import numpy as np
from scipy import stats

rates = np.array([0, 40, 80, 120, 160])
g, m = len(rates), 6
true_means = 6.8 - 2.7 * np.exp(-0.018 * rates)
rng = np.random.default_rng(1515)
group = np.repeat(np.arange(g), m)
```

kg N per hectare
5 rates, 6 plots per rate
unknown to the analyst

plot i received rate group[i]

```

y = np.round(true_means[group] + rng.normal(0, 0.5, g * m), 2)  # yield, t/ha
n = len(y)

n_k = np.bincount(group).astype(float)
means = np.bincount(group, weights=y) / n_k
nu = n - g
s = np.sqrt(np.sum((y - means[group]) ** 2) / nu)

def contrast(c, psi0=0.0, mult=None):
    """Estimate, standard error, t statistic, SS and interval for sum_k c_k mu_k."""
    est = c @ means
    v = np.sum(c ** 2 / n_k)  # Var(estimate) / sigma^2
    se = s * np.sqrt(v)
    mult = stats.t.ppf(0.975, nu) if mult is None else mult
    return est, se, (est - psi0) / se, est ** 2 / v, (est - mult * se, est + mult * se)

c_fert = np.array([-1, 0.25, 0.25, 0.25, 0.25])  # fertilized (average) minus control
c_top = np.array([0, 0, -1, 0, 1])  # 160 against 80 kg N/ha
for name, c in (("fertilized - control", c_fert), ("160 - 80", c_top)):
    est, se, t, ss, (lo, hi) = contrast(c)
    print(f"{name:21s} {est:6.3f} se {se:.3f} t = {t:5.2f} SS = {ss:6.3f}"
          f" 95% interval [{lo:.3f}, {hi:.3f}]"
    )

```

[Run this code online](#)

Each was a single planned question, so each gets an unadjusted t interval; joint coverage for the pair would use Bonferroni's $t_{25,0.0125}$. Writing the questions down in advance fixes a small family ([Section 13.6](#)).

15.2.4 Families of comparisons

Three families come up in almost every one-way analysis, and [Chapter 13](#) supplies a method for each. The multipliers below multiply the standard error of a difference of two means, $s\sqrt{2/m} = 0.3009$ here.

Comparisons with a control. The four differences $\mu_k - \mu_1$, $k = 2, \dots, 5$, form a planned family of size four. Bonferroni's method (Theorem 13.4.1) uses the multiplier $t_{25,0.05/8} = 2.692$ in place of the unadjusted 2.060:

Rate against control	Difference	Bonferroni 95% interval
40	1.232	[0.422, 2.041]
80	2.052	[1.242, 2.861]
120	2.510	[1.700, 3.320]
160	2.647	[1.837, 3.456]

Every fertilized rate beats the control. The four estimates share $\bar{y}_1$, so they are positively correlated, with correlation $\frac{1}{2}$ in a balanced design ([Exercise \(B4\)](#)). Dunnett (1955) computed the exact joint distribution of the four t statistics under this correlation; his multiplier here is 2.606 ([Exercise \(B2\)](#) characterizes it), a few percent shorter than Bonferroni's. We do not derive its tables.

All pairs. Tukey’s method (Theorem 13.3.2) covers all ten differences $\mu_k - \mu_l$ with exact joint coverage in a balanced design. With $q_{0.05}(5, 25) = 4.153$ the half-width is $q s / \sqrt{m} = 0.884$, and six of the ten pairs are declared different. The four that are not are 40 and 80, 80 and 120, 80 and 160, and 120 and 160. The conclusions are not transitive: 40 is not distinguished from 80, nor 80 from 160, yet 40 and 160 are declared different, since $6.652 - 5.237 = 1.415$ exceeds 0.884. “Not declared different” means “not enough evidence of a difference”, which is not an equivalence relation, so displays that sort treatments into groups of “equal” means should be read with care, especially in unbalanced designs.

Contrasts suggested by the data. Looking at Figure 15.1.1, one might ask whether the response has levelled off, by comparing the average of the two highest rates with 80 kg N/ha, $\frac{1}{2}(\mu_4 + \mu_5) - \mu_3$. The estimate is 0.527 with standard error 0.261. The unadjusted interval $[-0.010, 1.063]$ is not legitimate, because the contrast was chosen after seeing the means. Scheffé’s interval (Eq. 13.2.4), with multiplier $\sqrt{4F_{0.05}(4, 25)} = 3.322$, is $[-0.339, 1.392]$. It is valid however the contrast was found, and it says the question is open.

```
import itertools

k_ctrl = g - 1                                # four comparisons with the
control                                     control
bonf = stats.t.ppf(1 - 0.05 / (2 * k_ctrl), nu) # Bonferroni multiplier
se_pair = s * np.sqrt(2 / m)
for k in range(1, g):
    d = means[k] - means[0]
    print(f"{rates[k]:3d} - 0: {d:6.3f} Bonferroni interval"
          f" [{d - bonf * se_pair:.3f}, {d + bonf * se_pair:.3f}]")

q = stats.studentized_range.ppf(0.95, g, nu)    # q_{0.05}(g, nu)
hsd = q * s / np.sqrt(m)                       # Tukey half-width, all pairs
pairs = list(itertools.combinations(range(g), 2))
different = [(int(rates[k]), int(rates[l])) for k, l in pairs if abs(means[k] -
    means[l]) > hsd]
print(f"Tukey half-width {hsd:.3f}; pairs declared different: {len(different)} of {
    len(pairs)}")
print("not declared different:",
      [(int(rates[k]), int(rates[l])) for k, l in pairs if abs(means[k] - means[l])
      <= hsd])
```

[Run this code online](#)

15.2.5 Why planned contrasts are powerful

A single contrast is tested with one numerator degree of freedom, the overall test with $g - 1$. When the scientific question is sharp, the contrast is the better test, and the reason is geometric. Write $\mathbf{P}_{\mathcal{T}}$ for the projection onto the treatment space and

$$\gamma_{\mathcal{T}} = \frac{\|\mathbf{P}_{\mathcal{T}}\boldsymbol{\theta}\|^2}{\sigma^2} = \frac{1}{\sigma^2} \sum_{k=1}^g n_k (\mu_k - \bar{\mu})^2,$$

the squared length of the treatment component of the mean in units of σ^2 . (Section 15.3 shows that it is the noncentrality of the overall F test.)

Proposition 15.2.2. The noncentrality of a contrast

Let $\mathbf{c}$ be a contrast with vector $\mathbf{u}_{\mathbf{c}} \in \mathcal{T}$, and suppose $\mathbf{P}_{\mathcal{T}}\boldsymbol{\theta} \neq \mathbf{0}$. The noncentrality of the test of $\psi = 0$ in Proposition 15.2.1 is

$$\delta^2 = \frac{\psi^2}{\sigma^2 v_{\mathbf{c}}} = \gamma_{\mathcal{T}} \cos^2 \varphi,$$

where φ is the angle between $\mathbf{u}_{\mathbf{c}}$ and $\mathbf{P}_{\mathcal{T}}\boldsymbol{\theta}$. In particular $\delta^2 \leq \gamma_{\mathcal{T}}$, with equality iff $\mathbf{u}_{\mathbf{c}}$ is parallel to $\mathbf{P}_{\mathcal{T}}\boldsymbol{\theta}$, that is, iff $c_k \propto n_k(\mu_k - \bar{\mu})$.

Proof. By Proposition 8.7.2(a) applied to the mean vector, $\psi = \mathbf{u}_{\mathbf{c}}^{\top} \boldsymbol{\theta}$, and $v_{\mathbf{c}} = \|\mathbf{u}_{\mathbf{c}}\|^2$ by the computation in the proof of that result. Since $\mathbf{u}_{\mathbf{c}} \in \mathcal{T}$, $\mathbf{u}_{\mathbf{c}}^{\top} \boldsymbol{\theta} = \mathbf{u}_{\mathbf{c}}^{\top} \mathbf{P}_{\mathcal{T}}\boldsymbol{\theta}$. So

$$\delta^2 = \frac{(\mathbf{u}_{\mathbf{c}}^{\top} \mathbf{P}_{\mathcal{T}}\boldsymbol{\theta})^2}{\sigma^2 \|\mathbf{u}_{\mathbf{c}}\|^2} = \frac{\|\mathbf{P}_{\mathcal{T}}\boldsymbol{\theta}\|^2}{\sigma^2} \cdot \frac{(\mathbf{u}_{\mathbf{c}}^{\top} \mathbf{P}_{\mathcal{T}}\boldsymbol{\theta})^2}{\|\mathbf{u}_{\mathbf{c}}\|^2 \|\mathbf{P}_{\mathcal{T}}\boldsymbol{\theta}\|^2} = \gamma_{\mathcal{T}} \cos^2 \varphi$$

by Eq. 6.2.2. Equality in $\cos^2 \varphi \leq 1$ holds iff the two vectors are parallel. The vector $\mathbf{P}_{\mathcal{T}}\boldsymbol{\theta}$ has entry $\mu_k - \bar{\mu}$ in group k (Theorem 6.5.1), so it is $\mathbf{u}_{\mathbf{c}}$ for $c_k = n_k(\mu_k - \bar{\mu})$. ■

If the departure from equal means lies along the planned contrast, the contrast test gets all of the noncentrality on one degree of freedom, while the overall test spreads the same noncentrality over $g - 1$. For the nitrogen design, with 25 error degrees of freedom, the powers at level 0.05 are:

$\gamma_{\mathcal{T}}$, all along the contrast	4	8	12
contrast test, $F(1, 25)$	0.485	0.776	0.914
overall test, $F(4, 25)$	0.269	0.516	0.714

The price is paid when the question is wrong: a contrast orthogonal to $\mathbf{P}_{\mathcal{T}}\boldsymbol{\theta}$ has power α however large the treatment differences, and a direction chosen at random captures on average only $1/(g - 1)$ of $\gamma_{\mathcal{T}}$ (Exercise (C1)).

```
def power_f(gamma, q, nu, alpha=0.05):
    return stats.ncf.sf(stats.f.ppf(1 - alpha, q, nu), q, nu, gamma)

for gamma in (4.0, 8.0, 12.0):
    # all of it along the planned contrast
    print(f"gamma = {gamma:4.1f}: contrast test {power_f(gamma, 1, nu):.3f},"
          f" omnibus F test {power_f(gamma, g - 1, nu):.3f}")
```

[Run this code online](#)

15.2.6 Exercises

15.2.6.1 A. Check your understanding

Exercise. A1

In the nitrogen trial, compute by hand the estimate and standard error of $\frac{1}{2}(\mu_4 + \mu_5) - \frac{1}{2}(\mu_1 + \mu_2)$, the average of the two highest rates minus the average of the two lowest, from the rate means of [Section 15.1](#) and $s = 0.5211$.

Solution. The estimate is $\frac{1}{2}(6.515 + 6.652) - \frac{1}{2}(4.005 + 5.237) = 6.5835 - 4.621 = 1.9625$. With $m = 6$, $v_c = 4 \cdot \frac{1}{4}/6 = 1/6$, so the standard error is $0.5211/\sqrt{6} \approx 0.213$.

Exercise. A2

Show directly that the t statistic of [Proposition 15.2.1\(b\)](#) with $\psi_0 = 0$ is unchanged when $\mathbf{c}$ is multiplied by $a > 0$ and changes sign when $a < 0$. What happens to the confidence interval?

15.2.6.2 B. Practice

Exercise. B1

In a balanced layout with m observations per group, show that the noncentrality of the test of a contrast is $\delta^2 = m\psi^2/(\sigma^2 \sum_k c_k^2)$. How many observations per group are needed for the test of $\mu_5 - \mu_3$ in the nitrogen trial to have power 0.8 at level 0.05 when the true difference is 0.5 t/ha and $\sigma = 0.5$? (Use software for the noncentral F , or the normal approximation.)

Solution. Here $v_c = \sum_k c_k^2/m$, so $\delta^2 = \psi^2/(\sigma^2 v_c) = m\psi^2/(\sigma^2 \sum_k c_k^2)$. For $\mu_5 - \mu_3$, $\sum_k c_k^2 = 2$, $\psi/\sigma = 1$, and $\delta^2 = m/2$. The normal approximation asks for $\delta \approx z_{0.025} + z_{0.2} = 1.96 + 0.84 = 2.80$, so $m \approx 2 \cdot 2.80^2 \approx 15.7$. The exact noncentral F with $5(m-1)$ error degrees of freedom gives $m = 17$, since the t quantile exceeds the normal one. The trial's $m = 6$ is far short.

Exercise. B2

A single contrast $\mathbf{c}$ matters above all. For a fixed total n , show that $v_c = \sum_k c_k^2/n_k$ is minimized, over positive real n_k with $\sum_k n_k = n$, by $n_k \propto |c_k|$, with minimum $(\sum_k |c_k|)^2/n$. What does this say about how to allocate plots between two treatments, and between a control and the average of four treatments?

Solution. By the Cauchy–Schwarz inequality, $(\sum_k |c_k|)^2 = (\sum_k (|c_k|/\sqrt{n_k})\sqrt{n_k})^2 \leq \sum_k (c_k^2/n_k) \sum_k n_k = nv_c$, with equality iff $|c_k|/\sqrt{n_k} \propto \sqrt{n_k}$, that is, $n_k \propto |c_k|$. Groups with $c_k = 0$ get no observations. For $\mu_1 - \mu_2$, equal allocation is best. For $\mu_1 - \frac{1}{4}(\mu_2 + \dots + \mu_5)$, the control should get as many plots as the four treatments together, $n_1 = n/2$ and $n_k = n/8$ for the others.

Exercise. B3

In a comparison of $g - 1$ treatments with a control, suppose the control gets n_0 observations and each treatment gets n_t , with $n_0 + (g - 1)n_t = n$ fixed. Show that $\sum_k \text{Var}(\bar{Y}_k - \bar{Y}_0)$ is minimized when $n_0 = \sqrt{g - 1} n_t$. For $g = 5$ and $n = 30$, what allocation does this suggest?

Solution. The sum is $\sigma^2(g-1)(1/n_0 + 1/n_t)$. Minimizing $1/n_0 + 1/n_t$ subject to $n_0 + (g-1)n_t = n$ with a Lagrange multiplier gives $1/n_0^2 = \lambda$ and $1/n_t^2 = (g-1)\lambda$, so $n_0/n_t = \sqrt{g-1}$. For $g = 5$ the control should get twice as many observations as each treatment: $n_0 = 10$ and $n_t = 5$. Contrast [Exercise \(B2\)](#): for the average of the four comparisons the optimal control share is $n/2$, that is $n_0 = 4n_t$. The best allocation depends on whether the four comparisons or their average is the target.

Exercise. B4

With n_0 observations on the control and n_t on each treatment, show that the estimates $\bar{Y}_k - \bar{Y}_0$ and $\bar{Y}_l - \bar{Y}_0$ of two treatment-versus-control differences have correlation $n_t/(n_t + n_0)$. In a balanced design it is $\frac{1}{2}$.

Solution. The covariance is $\text{Var}(\bar{Y}_0) = \sigma^2/n_0$, since the treatment means are independent of each other and of $\bar{Y}_0$. Each variance is $\sigma^2(1/n_t + 1/n_0)$. The ratio is $(1/n_0)/(1/n_t + 1/n_0) = n_t/(n_t + n_0)$.

15.2.6.3 C. Going deeper**Exercise. C1**

Let $q = g - 1 \geq 2$ and let $\mathbf{u}$ be uniformly distributed on the unit sphere of the q -dimensional space $\mathcal{T}$ (for example $\mathbf{u} = \mathbf{w}/\|\mathbf{w}\|$ with $\mathbf{w} \sim \mathbf{N}$ on $\mathcal{T}$ with covariance $\mathbf{P}_{\mathcal{T}}$). Show that $\mathbb{E} \cos^2 \varphi = 1/q$, where φ is the angle between $\mathbf{u}$ and a fixed nonzero $\mathbf{v} \in \mathcal{T}$. Interpret this with [Proposition 15.2.2](#).

Solution. Choose an orthonormal basis of $\mathcal{T}$ whose first vector is $\mathbf{v}/\|\mathbf{v}\|$. In these coordinates $\cos^2 \varphi = u_1^2$. By symmetry the u_i^2 have a common mean, and they sum to 1, so $\mathbb{E} u_1^2 = 1/q$. By [Proposition 15.2.2](#), a contrast whose direction is unrelated to the truth captures on average a fraction $1/q$ of $\gamma_{\mathcal{T}}$ on its one degree of freedom, the same average share per degree of freedom as the overall test. Knowledge, not the count of degrees of freedom, is what makes a planned contrast more powerful.

Exercise. C2

The Scheffé multiplier for a family of contrasts depends on the dimension of the space the family spans (Theorem 13.2.2). Show that the $g - 1$ comparisons with a control span $\mathcal{T}$,

and that the family of the four “successive differences” $\mu_{k+1} - \mu_k$ does too. Why, then, are Bonferroni or Dunnett intervals shorter than Scheffé’s for these families?